2454.TW | 聯發科 MediaTek | Full investment artifact

Full active edition | Financial methods and portfolio Base overrides preserved.

Active approved edition: 30 September 2026. Framework 8.0.

Q3 2027 ceiling: TWD 4,888.5056 per share, valuing Q3 2027 through Q2 2028. **Analytical Base: 77/100. Protected Excel Base: 78/100.**

Valuation method: 20x parent economic operating profit. Shares in millions; TWD billions x 1000 / shares. Minority claims retained. Financial amounts: TWD B. Price, upside and ranking remain workbook-owned. This update adopts the approved proposal; it is not a new earnings release or a fresh stock-price quote.

Approved breadcrumb inputs and source images

These probabilities are explicit investment judgments under Framework 8.0, not measured statistical frequencies or claims of confirmed orders. Unconfirmed but credible evidence can affect the model when its company-specific transmission is quantified.

Goldman cloud ASIC revenue gap

Active probability: 30% like-for-like revenue and execution case.

The named-company forecast is materially above the current cloud-ASIC path. Use a 30% joint probability of comparable revenue scope and delivery. Program mapping corroborates the same opportunity and is not added separately.

Operating bridge: 2027/28 conditional ASIC sales = USD20.25B/52.5B x 32 TWD/USD, versus current USD14B/27.5B. Preserve quarterly mix. Incremental GP at 42%; incremental operating expense at the existing quarter opex/revenue ratio. Retain minority claims. After 2028 carry only the 2028 absolute annual uplift, without further growth.

Limits and counter-evidence: Revenue inclusion of HBM or other pass-through and calendar/fiscal alignment remain unresolved. 30% lies in the framework single-source band. Costs scale with new revenue; the old zero-incremental-opex sensitivity is superseded. 2029/30 continuation is our assumption.

Exhibit 3: Our bottom-up AI Infrastructure Capex build implies $1.3tn/$2tn in annual AI infrastructure spending on compute/networking in 2027/28

	2026	2027	2028
Nvidia			
Total Blackwell GW Deployments	12	3	-
Total Rubin GW Deployments	2	15	19
Total Feynman GW Deployments			3
Total GW deployments	**14**	**18**	**22**
CapEx per GW ($bn)	50	50	50
Total CapEx with Nvidia products ($bn)	**682**	**879**	**1,091**
Broadcom			
Custom XPU revenue ($mn)	39,373	86,176	168,126
AI Connectivity revenue ($mn)	18,465	39,300	72,180
Total AI Revenue ($mn)	57,838	125,477	240,306
Total GW deployments	**5**	**10**	**20**
CapEx per GW ($bn)	25	25	25
Total CapEx with Broadcom products ($bn)	**126**	**250**	**500**
AMD			
Datacenter GPU revenue ($mn)	15,662	42,317	56,017
Revenue per GW ($mn)	17,500	17,500	17,500
Total GW Deployments	**1**	**2**	**3**
CapEx per GW ($bn)	35	35	35
Total CapEx with AMD products ($bn)	**31**	**85**	**112**
MediaTek			
Datacenter ASIC revenue ($mn)	2,200	20,250	52,500
Revenue per GW ($mn)	5,000	5,000	5,000
Total GW Deployments	**0.4**	**4**	**11**
CapEx per GW ($bn)	25	25	25
Total CapEx with MediaTek products ($bn)	**11**	**101**	**263**
Marvell			
Datacenter ASIC revenue ($mn)	2,283	4,662	8,435
Revenue per GW ($mn)	5,000	5,000	5,000
Total GW Deployments	**0.5**	**0.9**	**2**
CapEx per GW ($bn)	25	25	25
Total CapEx with Marvell products ($bn)	**11**	**23**	**42**
Consolidated Summary			
Total GW Deployments	20	35	57
Total AI CapEx ($bn)	**861**	**1,338**	**2,008**
Implied CapEx/GW ($bn)	42	38	35

Source: Company data, Goldman Sachs Global Investment Research

User-supplied evidence for Goldman cloud ASIC revenue gap. The quoted forecast or estimate retains the source status described above.

Company	Chip	Process	Design service provider	AI card assembler	AI server assembler	Memory specs	Memory bandwidth	Mass produ
AWS								
	Inferentia 2	7nm TSMC	Alchip (back-end)	Accton	Wiwynn	HBM2e 32GB (2x16)	820 GB/s	2H22
	Inferentia 2.5 / Trainium 2	5nm TSMC	Marvell (back-end)	Accton	Wiwynn	HBM3 96GB (6x16)	2,900 GB/s	2H24
	Trainium 3	3nm TSMC	Alchip (back-end)	Accton/Fabrinet	Wiwynn			1H26
	Trainium 4	2nm TSMC	Alchip (back-end)	Accton/Fabrinet	Wiwynn			2027
	Graviton v3 (CPU)	5nm TSMC	AWS in-house		Wiwynn/Hon Hai			2022
	Graviton v4 (CPU)	4nm TSMC	AWS in-house		Wiwynn/Hon Hai			2024
Google								
	TPUv5p	7nm TSMC	Broadcom	Celestica	Celestica	HBM2e 95GB	2,765 GB/s	2023
	TPUv6e (Trillium)	5nm TSMC	Broadcom	Celestica	Celestica	HBM3/3e 32GB	1,600 GB/s	2024
	TPUv6p	3nm TSMC	Broadcom	Celestica	Celestica			2025
	TPUv7e	3nm TSMC	Mediatek (back-end & I/O)	Celestica	Celestica			2H25
	TPUv7p (Ironwood)	3nm TSMC	Broadcom	Celestica	Celestica			2H25
	TPUv8i	3nm TSMC	Broadcom	Celestica	Celestica			2H26
	TPUv8t	3nm TSMC	Mediatek (back-end & I/O)	Celestica	Celestica			2H26
	Axion (CPU)	5nm TSMC	Marvell (back-end)			DDR5 576GB		1H24
	Axion 2 (CPU)	3nm TSMC	GUC (wafer logistics mgnt)					2026
MSFT								
	Maia 100	5nm TSMC	GUC (wafer logistics mgnt)	n.a.		HBM2e 64GB (4x16)	1,800 GB/s	1H24
	Maia 200	3nm TSMC	Marvell (back-end)					2026
	Cobalt 100 (CPU)	5nm TSMC	GUC (wafer logistics mgnt)			DDR5 12 channel		2H24
	Cobalt 200 (CPU)	3nm TSMC	GUC (wafer logistics mgnt)					2026
Meta								
	MTIA	7nm TSMC	Broadcom	n.a.	Wiwynn	LPDDR5 128GB		2H23-1H24
	MTIA v2	5nm TSMC	Broadcom		Wiwynn	LPDDR5 128GB	204.8 GB/s	2H25
	MTIA v3/MTIA T-V1	3nm TSMC	Broadcom		Quanta / Celestica	HBM3E		2H26
	(CPU gen 1)	3nm TSMC	SocioNext					2026
Intel (Habana)								
	Gaudi 2	7nm TSMC	Alchip (back-end)	Accton	Wiwynn	HBM2E 96GB (6x16)	2,450 GB/s	2H23
	Gaudi 3	5nm TSMC	Alchip (back-end)	Accton	Wiwynn	HBM2E 128GB (8x16)	3,700 GB/s	2H24
	Falcon shores							
	Jaguar Shores	Intel 18A						

User-supplied evidence for Goldman cloud ASIC revenue gap. The quoted forecast or estimate retains the source status described above.

Sources: GS, ASIC.

Transmission field	Active treatment
Probability	30% like-for-like revenue and execution case
Recognition	2027/28 conditional ASIC sales = USD20.25B/52.5B x 32 TWD/USD, versus current USD14B/27.5B. Preserve quarterly mix. Incremental GP at 42%; incremental operating expense at the existing quarter opex/revenue ratio. Retain minority claims. After 2028 carry only the 2028 absolute annual uplift, without further growth.
Overlap / uncertainty	Revenue inclusion of HBM or other pass-through and calendar/fiscal alignment remain unresolved. 30% lies in the framework single-source band. Costs scale with new revenue; the old zero-incremental-opex sensitivity is superseded. 2029/30 continuation is our assumption.
Raise probability	Timely qualifying milestones, comparable revenue scope, stronger customer allocation and observed unit economics.
Lower probability	Delay, customer substitution, weaker realized economics, overlap or evidence contradicting the stated source scope.

1. Supercycle, supply chain and durable advantage

MediaTek can reuse system design, high-speed interfaces, advanced-node integration and supply-chain relationships across mobile, edge and custom cloud accelerators. Larger accelerator dies and complex packages increase its dollar content. The company competes on performance per watt and total system cost; customers retain architectural control and can use competing suppliers. The opportunity expands beyond phones without making every cloud component part of MediaTek revenue.

Customer architecture → MediaTek design/IP/integration → TSMC wafers and qualified packaging → customer accelerator/system. MediaTek supports both CoWoS and EMIB-T; supplying an accelerator does not imply owning the foundry, HBM or the entire rack. Edge computing combines merchant SoCs, connectivity and power management, with shared R&D across products.

Business / curve	Stage	Evidence and next inflection
Smartphone	Mature with flagship mix upgrade	Weak units coexist with richer 2nm products.
Smart Edge	Scaling, qualitative 25-50% band	Connectivity, computing and auto projects add share.
Cloud accelerators	Initial production to rapid scaling, 5-20% band	First project Q4 2026; second early 2028.
Optical/system platforms	Development, below 10% band	No independent commercial value sized in the base.

Stage percentages are qualitative framework bands relative to a possible mature opportunity, not measured market penetration. A current capacity constraint can coexist with a larger long-term TAM; qualification and economic capture decide how much reaches this company.

2. Forward TAM and expansion

The management 2027 accelerator SAM excludes HBM. The approved 30% alternative forecast modifies only cloud ASIC revenue; it does not add whole-server capex or a separate second supplier-map opportunity. Mobile, edge and Power IC paths retain their original economics.

Year	Current ASIC revenue TWD B	Current ASIC USD B at 32	Treatment
2027	508.000	15.875	Weighted forecast; not booked orders
2028	1,120.000	35.000	Weighted forecast; not booked orders
2029	1,449.600	45.300	Weighted forecast; not booked orders
2030	1,808.000	56.500	Weighted forecast; not booked orders

Phase 1 deployment, Phase 2 wider platform adoption and Phase 3 new edge applications retain their 100%/50%/20% thematic weights. These are not another multiplier on the company forecast.

3. Current earnings and management evidence

Q2 revenue was NT$152.183B, GP 70.308B and OP 22.868B. R&D was 39.361B, selling 4.685B and administration 3.395B, with total opex 47.440B after rounding. Q1 revenue/OP were 149.151B/22.891B. Q2 sales proportions were mobile 41%, Smart Edge 53% and Power IC 6%. July and August unaudited revenue of 48.475B and 64.183B provide a newer quarterly check than the July outlook. [S1/S2/S5]

Both official April 30 and July 31 transcripts were read through Q&A. The accelerator opportunity moved from a 2027 SAM of US$70-80B and 10-15% share to US$80B and 15-20% share. Management explicitly excludes HBM from this SAM. The first project starts production in Q4 2026; full-year data-center revenue should exceed US$2B. The second project now points to early 2028 production. Management expects similar gross margins between generations, slightly below corporate GM, with substantial operating leverage; it did not provide a numeric future OP margin. [S3/S4]

Latest two issuer transcripts, current financials and monthly sales are available. Customer identity, project codenames, exact chip ASPs, packaging unit costs and 2028-30 market size are not publicly specified in those sources. Model allocations remain LOW confidence where not directly guided.

4. Local evidence and calibration packet

Native Chinese prepared remarks and English Q&A in the official PDF were cross-checked. July plus August sales total NT$112.658B. Model Q3 170B requires September 57.342B, above July but below August. This exceeds the July guidance high end 159.8B by 6.4%, explicitly an analyst revision based on later monthly evidence. Category allocation is not reported. AH12 procurement evidence is the customer production schedule, secured capacity commentary and the US$5B discretionary financing budget; the latter is neither a signed order nor equity already issued.

5. Summing thesis score

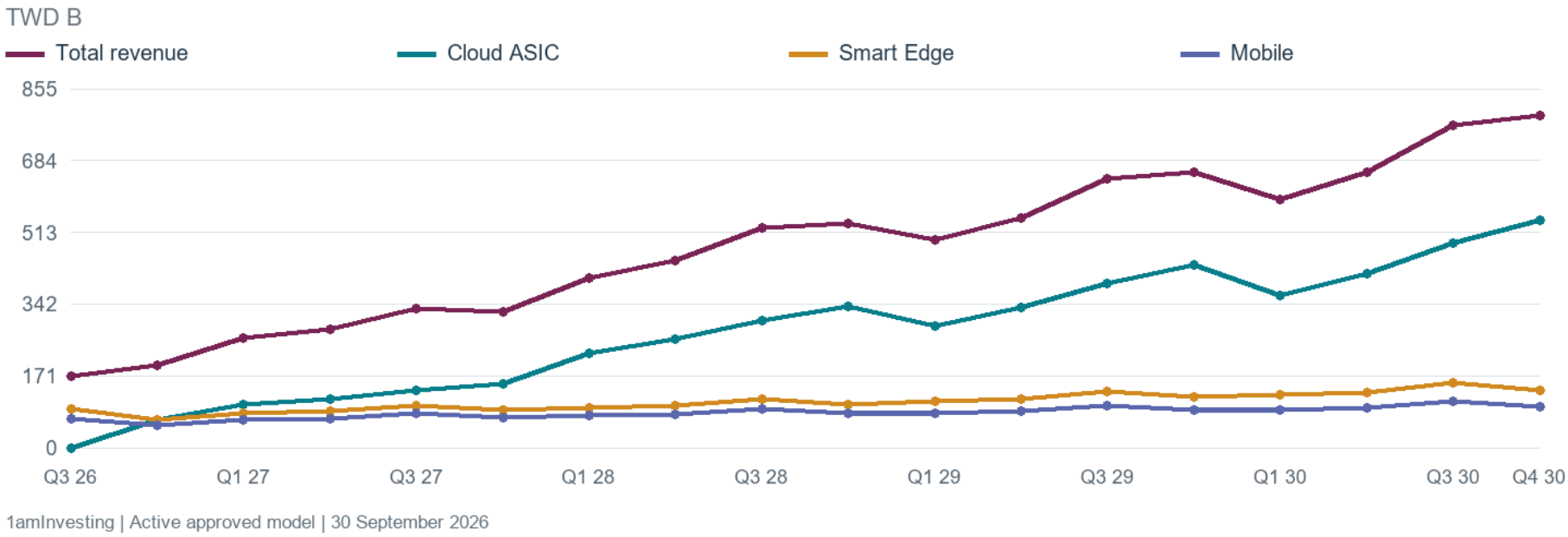
Category	Points / maximum	Evidence
Core Thesis	16/20	Strong integration and ecosystem; customer concentration remains.
Cycle Stage	9/10	First accelerator production near; large next-year ramp remains execution dependent.
TAM Expansion	10/10	Issuer raised accelerator capture target with HBM boundary explicit.
Catalysts	5/5	Q4 first project, early-2028 second project and edge products.
Macro	8/10	Cloud AI demand strong; memory inflation pressures consumer units.
Alpha	3/5	Gross versus operating-margin economics and net revenue boundaries matter.
Risks	-5	Five distinct classes.

Sum: 46/60 × 100 = 76.666667%, conventionally rounded to 77%. This advisory thesis score does not replace any portfolio Base override. The separate risk classes below total the displayed deduction; holding status and classification do not waive that deduction.

Revenue vectors and unit economics

Revenue uses historical anchors times explicit quarterly conversion factors. Mobile GM is 48%; Smart Edge GM is 45.5% after Q3 2026; Power IC is 40%; cloud ASIC is 42%. These are model estimates, not segment margin guidance. Cloud ASIC excludes a second HBM, rack, CPO, SerDes or NRE add-on.

Quarterly revenue and mix



1amInvesting | Active approved model | 30 September 2026

Active current quarterly model. Source: 2454_TW_Q_Model. Values and units are shown in the table below.

Quarter	Mobile	Smart Edge	Power IC	Cloud ASIC	Total revenue
Q3 2026	68.000	92.000	10.000	0.000	170.000
Q4 2026	54.000	65.000	9.000	67.200	195.200
Q1 2027	66.700	82.800	9.200	101.600	260.300
Q2 2027	69.600	86.400	9.600	116.840	282.440
Q3 2027	81.200	100.800	11.200	137.160	330.360
Q4 2027	72.500	90.000	10.000	152.400	324.900
Q1 2028	75.900	95.220	10.350	224.000	405.470
Q2 2028	79.200	99.360	10.800	257.600	446.960
Q3 2028	92.400	115.920	12.600	302.400	523.320

Quarter	Mobile	Smart Edge	Power IC	Cloud ASIC	Total revenue
Q4 2028	82.500	103.500	11.250	336.000	533.250
Q1 2029	82.800	110.400	11.730	289.920	494.850
Q2 2029	86.400	115.200	12.240	333.408	547.248
Q3 2029	100.800	134.400	14.280	391.392	640.872
Q4 2029	90.000	120.000	12.750	434.880	657.630
Q1 2030	89.700	126.500	13.340	361.600	591.140
Q2 2030	93.600	132.000	13.920	415.840	655.360
Q3 2030	109.200	154.000	16.240	488.160	767.600
Q4 2030	97.500	137.500	14.500	542.400	791.900

Mobile operating inputs

Quarter	Anchor	Conversion factor	Revenue	GM	Gross profit
Q3 2026	62.395	1.090	68.000	0.480	32.640
Q4 2026	62.395	0.865	54.000	0.480	25.920
Q1 2027	62.395	1.069	66.700	0.480	32.016
Q2 2027	62.395	1.115	69.600	0.480	33.408
Q3 2027	62.395	1.301	81.200	0.480	38.976
Q4 2027	62.395	1.162	72.500	0.480	34.800
Q1 2028	62.395	1.216	75.900	0.480	36.432
Q2 2028	62.395	1.269	79.200	0.480	38.016
Q3 2028	62.395	1.481	92.400	0.480	44.352
Q4 2028	62.395	1.322	82.500	0.480	39.600
Q1 2029	62.395	1.327	82.800	0.480	39.744
Q2 2029	62.395	1.385	86.400	0.480	41.472
Q3 2029	62.395	1.616	100.800	0.480	48.384
Q4 2029	62.395	1.442	90.000	0.480	43.200
Q1 2030	62.395	1.438	89.700	0.480	43.056
Q2 2030	62.395	1.500	93.600	0.480	44.928
Q3 2030	62.395	1.750	109.200	0.480	52.416
Q4 2030	62.395	1.563	97.500	0.480	46.800

Smart Edge operating inputs

Quarter	Anchor	Conversion factor	Revenue	GM	Gross profit
Q3 2026	80.657	1.141	92.000	0.452	41.560
Q4 2026	80.657	0.806	65.000	0.455	29.575
Q1 2027	80.657	1.027	82.800	0.455	37.674
Q2 2027	80.657	1.071	86.400	0.455	39.312
Q3 2027	80.657	1.250	100.800	0.455	45.864
Q4 2027	80.657	1.116	90.000	0.455	40.950
Q1 2028	80.657	1.181	95.220	0.455	43.325
Q2 2028	80.657	1.232	99.360	0.455	45.209
Q3 2028	80.657	1.437	115.920	0.455	52.744
Q4 2028	80.657	1.283	103.500	0.455	47.093
Q1 2029	80.657	1.369	110.400	0.455	50.232
Q2 2029	80.657	1.428	115.200	0.455	52.416
Q3 2029	80.657	1.666	134.400	0.455	61.152
Q4 2029	80.657	1.488	120.000	0.455	54.600
Q1 2030	80.657	1.568	126.500	0.455	57.558
Q2 2030	80.657	1.637	132.000	0.455	60.060
Q3 2030	80.657	1.909	154.000	0.455	70.070
Q4 2030	80.657	1.705	137.500	0.455	62.562

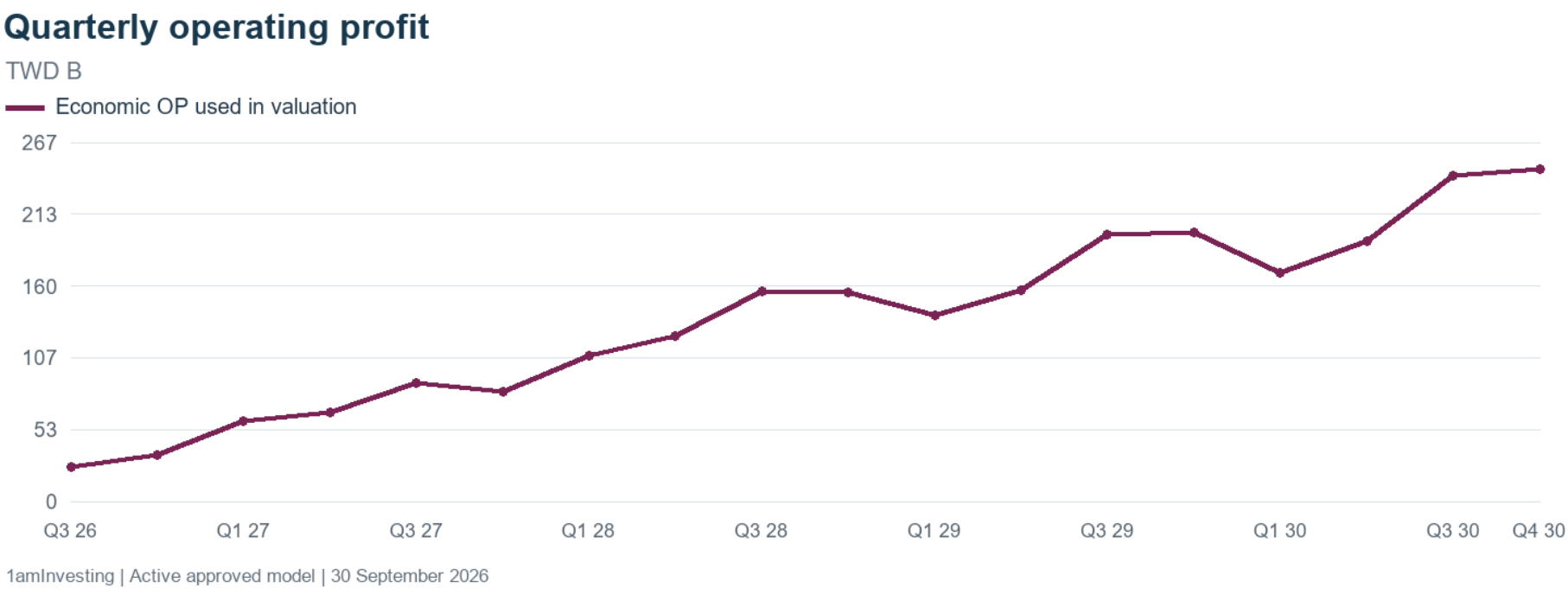
Power IC operating inputs

Quarter	Anchor	Conversion factor	Revenue	GM	Gross profit
Q3 2026	9.131	1.095	10.000	0.400	4.000
Q4 2026	9.131	0.986	9.000	0.400	3.600
Q1 2027	9.131	1.008	9.200	0.400	3.680
Q2 2027	9.131	1.051	9.600	0.400	3.840
Q3 2027	9.131	1.227	11.200	0.400	4.480
Q4 2027	9.131	1.095	10.000	0.400	4.000
Q1 2028	9.131	1.134	10.350	0.400	4.140
Q2 2028	9.131	1.183	10.800	0.400	4.320
Q3 2028	9.131	1.380	12.600	0.400	5.040
Q4 2028	9.131	1.232	11.250	0.400	4.500
Q1 2029	9.131	1.285	11.730	0.400	4.692
Q2 2029	9.131	1.340	12.240	0.400	4.896
Q3 2029	9.131	1.564	14.280	0.400	5.712
Q4 2029	9.131	1.396	12.750	0.400	5.100
Q1 2030	9.131	1.461	13.340	0.400	5.336
Q2 2030	9.131	1.524	13.920	0.400	5.568
Q3 2030	9.131	1.779	16.240	0.400	6.496
Q4 2030	9.131	1.588	14.500	0.400	5.800

Cloud ASIC operating inputs

Quarter	Anchor	Conversion factor	Revenue	GM	Gross profit
Q3 2026	1.000	0.000	0.000	0.420	0.000
Q4 2026	1.000	67.200	67.200	0.420	28.224
Q1 2027	1.000	101.600	101.600	0.420	42.672
Q2 2027	1.000	116.840	116.840	0.420	49.073
Q3 2027	1.000	137.160	137.160	0.420	57.607
Q4 2027	1.000	152.400	152.400	0.420	64.008
Q1 2028	1.000	224.000	224.000	0.420	94.080
Q2 2028	1.000	257.600	257.600	0.420	108.192
Q3 2028	1.000	302.400	302.400	0.420	127.008
Q4 2028	1.000	336.000	336.000	0.420	141.120
Q1 2029	1.000	289.920	289.920	0.420	121.766
Q2 2029	1.000	333.408	333.408	0.420	140.031
Q3 2029	1.000	391.392	391.392	0.420	164.385
Q4 2029	1.000	434.880	434.880	0.420	182.650
Q1 2030	1.000	361.600	361.600	0.420	151.872
Q2 2030	1.000	415.840	415.840	0.420	174.653
Q3 2030	1.000	488.160	488.160	0.420	205.027
Q4 2030	1.000	542.400	542.400	0.420	227.808

Operating profit and cost bridge



Current operating-profit path. Consolidated and parent economic profit are distinguished where minority claims apply.

Quarter	Status	Revenue	Gross profit	Opex	Consolidated OP	Valuation OP	OP / revenue
Q3 2026	Model	170.000	78.200	52.700	25.500	25.200	14.8%
Q4 2026	Model	195.200	87.319	52.500	34.819	34.519	17.7%
Q1 2027	Model	260.300	116.042	56.358	59.684	59.384	22.8%
Q2 2027	Model	282.440	125.633	58.877	66.756	66.456	23.5%
Q3 2027	Model	330.360	146.927	58.888	88.039	87.739	26.6%
Q4 2027	Model	324.900	143.758	61.656	82.102	81.802	25.2%
Q1 2028	Model	405.470	177.977	69.418	108.559	108.209	26.7%
Q2 2028	Model	446.960	195.737	72.733	123.004	122.654	27.4%

Quarter	Status	Revenue	Gross profit	Opex	Consolidated OP	Valuation OP	OP / revenue
Q3 2028	Model	523.320	229.144	72.759	156.384	156.034	29.8%
Q4 2028	Model	533.250	232.312	76.649	155.663	155.313	29.1%
Q1 2029	Model	494.850	216.434	77.608	138.827	138.427	28.0%
Q2 2029	Model	547.248	238.815	81.189	157.626	157.226	28.7%
Q3 2029	Model	640.872	279.633	81.211	198.421	198.021	30.9%
Q4 2029	Model	657.630	285.550	85.254	200.296	199.896	30.4%
Q1 2030	Model	591.140	257.822	87.244	170.577	170.077	28.8%
Q2 2030	Model	655.360	285.209	91.180	194.029	193.529	29.5%
Q3 2030	Model	767.600	334.009	91.199	242.810	242.310	31.6%
Q4 2030	Model	791.900	342.971	95.525	247.445	246.945	31.2%

Parent economic OP deducts the unchanged minority proxy: TWD0.3B per quarter through 2027, 0.35B in 2028, 0.4B in 2029 and 0.5B in 2030.

8. Multiple, shares and recognition convention

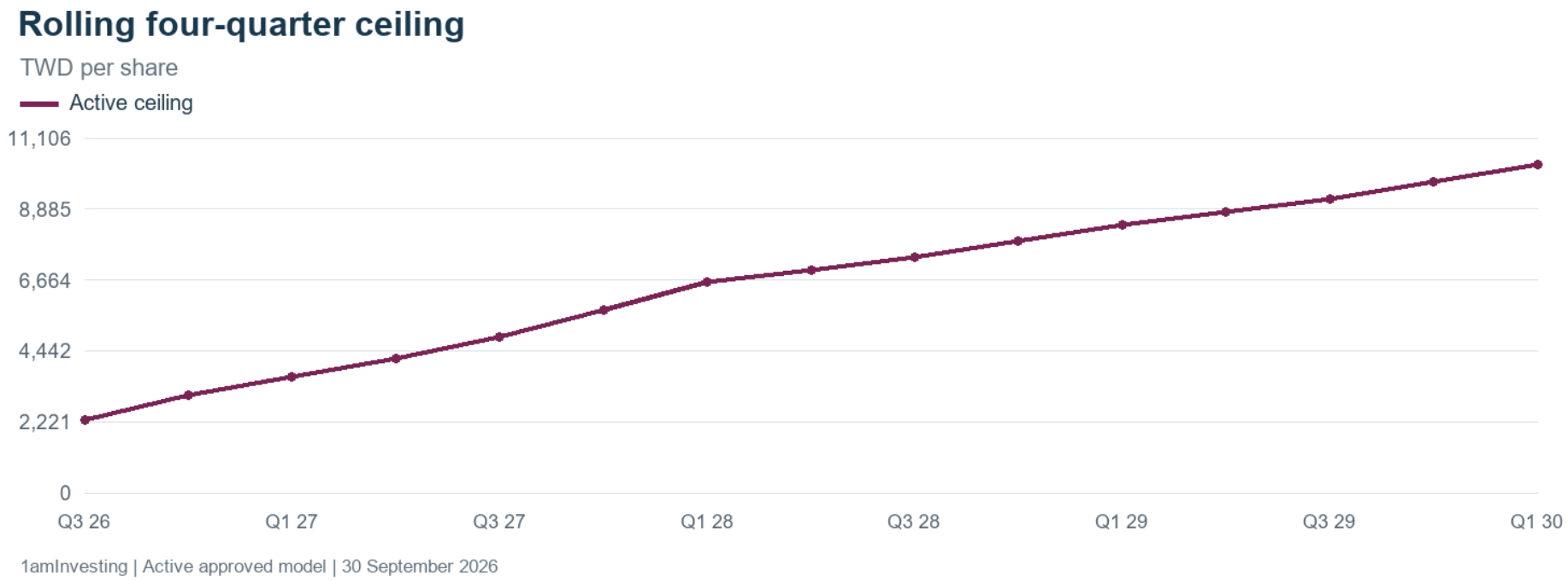
Retain 20× rolling four-quarter operating earnings. It is an OP multiple, not an after-tax P/E. Stronger profit is generated by explicit revenue and cost inputs, without increasing the valuation multiple.

Retain 1,630M forward share proxy, compared with approximately 1,603.9M issued shares inferred from NT$16.039B capital at NT$10 par. The 26.1M allowance is an assumption for equity awards, not a claimed audited fully diluted count. Add 0.5% annual dilution. The US$5B financing authorization permits financing flexibility; its final debt/equity mix is unknown and is not treated as an already completed equity issue.

The valuation denominator is the ceiling quarter’s diluted-share proxy. It is not a historical EPS weighted-average assumption disguised as a fully known future capitalization. Unit factor: 1000; this converts the model units to TWD per share. No cash balance, enterprise-value adjustment or after-tax earnings substitution is added to the operating-profit ceiling.

Rolling forward-four-quarter ceiling

The ceiling uses operating profit in the stated quarter plus the next three quarters. The company-specific method remains **20x parent economic operating profit**. Shares in millions; TWD billions x 1000 / shares. Minority claims retained. Probability is applied to the specified operating increment once; the ceiling receives no second probability or Base multiplier.



Current ceiling path. A negative-profit window is marked neg rather than assigned a positive operating-profit ceiling.

Quarter	Exact earnings window	Forward 4Q OP	Multiple	Shares millions	Unit / FX	Ceiling TWD
Q3 2026	Q3 2026 to Q2 2027	185.559	20x	1,630.000000	1,000.000	2,276.7968
Q4 2026	Q4 2026 to Q3 2027	248.098	20x	1,632.033691	1,000.000	3,040.3593
Q1 2027	Q1 2027 to Q4 2027	295.382	20x	1,634.069919	1,000.000	3,615.2872
Q2 2027	Q2 2027 to Q1 2028	344.207	20x	1,636.108688	1,000.000	4,207.6312
Q3 2027	Q3 2027 to Q2 2028	400.405	20x	1,638.150000	1,000.000	4,888.5056
Q4 2027	Q4 2027 to Q3 2028	468.700	20x	1,640.193859	1,000.000	5,715.1775
Q1 2028	Q1 2028 to Q4 2028	542.211	20x	1,642.240269	1,000.000	6,603.3092
Q2 2028	Q2 2028 to Q1 2029	572.428	20x	1,644.289231	1,000.000	6,962.6215
Q3 2028	Q3 2028 to Q2 2029	607.000	20x	1,646.340750	1,000.000	7,373.9271

Quarter	Exact earnings window	Forward 4Q OP	Multiple	Shares millions	Unit / FX	Ceiling TWD
Q4 2028	Q4 2028 to Q3 2029	648.987	20x	1,648.394829	1,000.000	7,874.1671
Q1 2029	Q1 2029 to Q4 2029	693.569	20x	1,650.451470	1,000.000	8,404.6002
Q2 2029	Q2 2029 to Q1 2030	725.220	20x	1,652.510677	1,000.000	8,777.1910
Q3 2029	Q3 2029 to Q2 2030	761.523	20x	1,654.572454	1,000.000	9,205.0732
Q4 2029	Q4 2029 to Q3 2030	805.812	20x	1,656.636803	1,000.000	9,728.2897
Q1 2030	Q1 2030 to Q4 2030	852.862	20x	1,658.703727	1,000.000	10,283.4734

Arithmetic and denominator sensitivity

- Q3 2026: (25.2000 + 34.5190 + 59.3839 + 66.4561) x 20 x 1000 / 1,630.000000 = 2,276.7968 TWD/share.
- Q4 2026: (34.5190 + 59.3839 + 66.4561 + 87.7395) x 20 x 1000 / 1,632.033691 = 3,040.3593 TWD/share.
- Q1 2027: (59.3839 + 66.4561 + 87.7395 + 81.8022) x 20 x 1000 / 1,634.069919 = 3,615.2872 TWD/share.
- Q2 2027: (66.4561 + 87.7395 + 81.8022 + 108.2093) x 20 x 1000 / 1,636.108688 = 4,207.6312 TWD/share.
- Q3 2027: (87.7395 + 81.8022 + 108.2093 + 122.6543) x 20 x 1000 / 1,638.150000 = 4,888.5056 TWD/share.
- Q4 2027: (81.8022 + 108.2093 + 122.6543 + 156.0342) x 20 x 1000 / 1,640.193859 = 5,715.1775 TWD/share.
- Q1 2028: (108.2093 + 122.6543 + 156.0342 + 155.3132) x 20 x 1000 / 1,642.240269 = 6,603.3092 TWD/share.
- Q2 2028: (122.6543 + 156.0342 + 155.3132 + 138.4265) x 20 x 1000 / 1,644.289231 = 6,962.6215 TWD/share.
- Q3 2028: (156.0342 + 155.3132 + 138.4265 + 157.2259) x 20 x 1000 / 1,646.340750 = 7,373.9271 TWD/share.
- Q4 2028: (155.3132 + 138.4265 + 157.2259 + 198.0212) x 20 x 1000 / 1,648.394829 = 7,874.1671 TWD/share.
- Q1 2029: (138.4265 + 157.2259 + 198.0212 + 199.8957) x 20 x 1000 / 1,650.451470 = 8,404.6002 TWD/share.
- Q2 2029: (157.2259 + 198.0212 + 199.8957 + 170.0774) x 20 x 1000 / 1,652.510677 = 8,777.1910 TWD/share.
- Q3 2029: (198.0212 + 199.8957 + 170.0774 + 193.5288) x 20 x 1000 / 1,654.572454 = 9,205.0732 TWD/share.
- Q4 2029: (199.8957 + 170.0774 + 193.5288 + 242.3103) x 20 x 1000 / 1,656.636803 = 9,728.2897 TWD/share.
- Q1 2030: (170.0774 + 193.5288 + 242.3103 + 246.9453) x 20 x 1000 / 1,658.703727 = 10,283.4734 TWD/share.

Q3 2027 denominator case	Shares millions	Ceiling TWD
10% fewer shares	1,474.335000	5,431.6729
Active schedule	1,638.150000	4,888.5056

Q3 2027 denominator case	Shares millions	Ceiling TWD
10% more shares	1,801.965000	4,444.0960

These are arithmetic sensitivities with unchanged profit, not extra operating scenarios or forecasts of financing.

Annual cross-check and demand scale

Calendar buckets sum the stated quarter labels; fiscal proxy mappings remain as in the company earnings section. These totals do not replace rolling valuation windows. 2026 is only H2 in this table.

Year	Revenue TWD B	Valuation OP TWD B	OP margin
2026 H2	365.200	59.719	16.4%
2027	1,198.000	295.382	24.7%
2028	1,909.000	542.211	28.4%
2029	2,340.600	693.569	29.6%
2030	2,806.000	852.862	30.4%

Operating driver and probability transmission



Difference attributable solely to the approved breadcrumb inputs. Negative values are retained; this is not an additional revenue pool.

Quarter	Revenue increment TWD B	OP increment TWD B
Q3 2026	0.0000	0.0000
Q4 2026	0.0000	0.0000
Q1 2027	12.0000	2.4419
Q2 2027	13.8000	2.9193
Q3 2027	16.2000	3.9163
Q4 2027	18.0000	4.1442
Q1 2028	48.0000	11.9422
Q2 2028	55.2000	14.2015

Quarter	Revenue increment TWD B	OP increment TWD B
Q3 2028	64.8000	18.2066
Q4 2028	72.0000	19.8907
Q1 2029	48.0000	12.6321
Q2 2029	55.2000	14.9946
Q3 2029	64.8000	19.0045
Q4 2029	72.0000	20.9061
Q1 2030	48.0000	13.0759
Q2 2030	55.2000	15.5040
Q3 2030	64.8000	19.5171
Q4 2030	72.0000	21.5548

12. Programme and probability ledger

Programme / signal	Source and probability	Amount / unit base	Treatment in this model	Escalation / recognition
First accelerator	Company production plan; 85% timing judgment	Q4 US$2.1B model versus >US$2B annual outlook	Included once in cloud vector	Check recognized deliveries, acceptance and cash conversion.
2027 share target	Management target, not guaranteed orders	US$80B ×17.5%=US$14B	Reference midpoint; active 30% comparable-scope increment is specified in the approved ledger	Confirm capacity and customer schedules.
Second accelerator	Design/packaging progress; 75% timing judgment	Early 2028 initial production	Embedded in 2028 SAM/share; no extra late-2027 sales	Track tape-out, qualified yield and first shipments.
Additional customers and rack solutions	Engagement; 40% conversion judgment	No disclosed amounts	Unquantified optionality; no rack revenue add-on	Signed scope and revenue ownership first.
CPO/CPC and 448G	Technology roadmap; 50% adoption judgment	448G readiness anticipated H2 2027; optical later	Contained within future product economics	Commercial qualification, not test-chip existence, drives capture.

Probabilities are analyst judgments under the framework signal hierarchy. Embedded revenue is not multiplied or added a second time. Where no defensible amount or unit base exists, the opportunity remains explicitly unquantified; confidence alone cannot create a revenue number. New evidence can enlarge the TAM, improve capture or move recognition earlier, but each requires an identifiable model input.

13. Scored risk classes

Risk class	Scored deduction	Mechanism and consequence
Customer concentration	-1	A dominant cloud programme can change volume or architecture.
Competition and customer-owned design	-1	Broadcom, other ASIC partners and internal design can reduce capture.
Supply and packaging execution	-1	Foundry, substrates and advanced packaging can constrain production.
Margin, cash and funding	-1	Inventory and supplier commitments precede revenue; business-model changes can dilute economics.
Consumer and geopolitical demand	-1	Phone weakness, regional trade risk and foundry concentration persist.

14. Dated catalysts, counters and escalation

1. **October 2026 monthly release - September sales** Validate the 57.342B monthly assumption behind Q3 170B.
2. **Q4 2026 - First ASIC production** Test >US$2B yearly contribution and working-capital release.
3. **2027 - Market share and opex leverage** Track capacity, customer shipments and spending dollars.
4. **Early 2028 - Second project** Production and yield replace inferred timing.
5. **Quarterly - Funding and cash** H1 OCF 6.594B was below PPE purchases 10.815B and intangible purchases 9.926B; profit is not immediate free cash.

15. Cross-book monitoring

Intel’s packaging opportunity and MediaTek’s ASIC revenue are different economic layers; do not assume MediaTek uses Intel wafer fabrication from EMIB-T support. Broadcom competes for accelerator work. Memory suppliers benefit from HBM content even though HBM is excluded from the company SAM. No separate changes to those portfolio artifacts are made.

16. Source register and model limits

S1 - Q2 financial statements (2026-07-31). Reported financial calibration and capital.

S2 - Q2 results release (2026-07-31). Earnings and reporting basis.

S3 - Official Q1 full call transcript (2026-04-30). Prior outlook, technology and business-model boundaries.

S4 - Official Q2 full call transcript (2026-07-31). Raised SAM/share, margins, schedule and financing budget.

S5 - Issuer monthly revenue register (2026-09-10). July/August unaudited monthly sales.

The customer remains an unnamed major US CSP in official guidance. Google project codenames, exclusive future awards and exact $18,000 chip ASPs in the old artifact are not treated as issuer-confirmed inputs. 2028-30 SAM and capture are extrapolations. Expanded TAM does not imply automatic revenue from every system component.

Workbook reference: v3_2_master_portfolio.xlsx, tab 2454_TW_Q_Model. The workbook records this approved research update; approved operating inputs and their dependent ceilings are synchronized; valuation methodology is unchanged.

Model reconciliation and limitations

Current tables match 2454_TW_Q_Model and the approved proposal at every available quarter. Excel Base overrides, valuation methods, share schedules and units are preserved. The next 12-16 months deserve the highest forecast weight; outer-year estimates are substantially less reliable.

Revisit probabilities and operating inputs as evidence changes. Official confirmation is not required to value a quantified breadcrumb; source uncertainty remains explicit.